Pedro Marco Achanccaray Diaz

Researcher and Lecturer at IGP/TU Braunschweig

Experienced researcher proficient in machine learning, deep learning, computer vision, and remote sensing, with expertise in applying these techniques to diverse fields such as agriculture mapping, oil and gas exploration/monitoring/extraction, and building heritage preservation. Exhibits competence in the supervision and orientation of master's and doctoral students during lectures and guiding them through the development of projects.

Contact Information

Bienroder Weg 81, Room 012
Institute of Geodesy and Photogrammetry – IGP
Technical University of Braunschweig – TUBS
Braunschweig, Germany

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Research Interests

Computer vision, Machine learning, Deep learning, Remote sensing

Education

Pontifical Catholic University of Rio de Janeiro (PUC-Rio)
Ph.D. in Electrical Engineering, 2019

Pontifical Catholic University of Rio de Janeiro (PUC-Rio)
M.Sc. in Electrical Engineering, 2014

National University of Engineering (UNI)
B.Sc. in Mechanical-Electrical Engineering, 2010

Professional/ Research Experience

Institute of Geodesy and Photogrammetry - IGP 2021-2028
Technische Universität Braunschweig – TUBS, Germany
Researcher and Lecturer
Research on projects using machine/deep learning methods and remote sensing data for different applications such as building heritage preservation, concrete damage detection, land use/land cover segmentation, 3D concrete printing quality assessment, and microplastic monitoring in rivers. Lecturer on the Deep Learning course in the Master program, and supervision of master and doctoral students.

Applied Computational Intelligence Lab – ICA 2019-2021
Pontifical Catholic University of Rio de Janeiro – PUC-Rio, Brazil
Researcher and Developer
Research and development of projects using machine/ deep learning methods and data from different domains (sea surface satellite images, sea floor seismic data, and ROV pipeline inspection videos) in the exploration, extraction, and monitoring tasks in the Oil and Gas industry. Supervisor of master and doctoral students.

Computer Vision Lab - LVC 2015-2018
Pontifical Catholic University of Rio de Janeiro – PUC-Rio, Brazil
Researcher
Development of public benchmarks for agricultural applications. My activities involved the pre-processing of sequences of multitemporal Sentinel-1 (SAR) images for agricultural monitoring in two municipalities in Brazil: **Campo Verde** and Luis Eduardo Magalhães (**LEM**). These projects were in cooperation with the National Institute for Space Research – INPE. LEM received financial support from the ISPRS Scientific Initiatives.

Teaching Experience

- Deep learning (IGP/TUBS), *Lecturer – Master program* 2022-2026
- Applied Generative AI and Computer Vision (PUCP), *Lecturer – Specialization course* 2026
- Deep learning for social sciences and public administration (PUCP), *Lecturer – Specialization course* 2022-2026
- Machine learning for social sciences (PUCP), *Lecturer – Specialization course* 2021

Achievements, Honors and Awards

- TU Braunschweig Lehrpreis 2024: Best international lecture, Deep Learning 2024
- Scholarship *Science without Borders* from *CAPES,* for visiting researcher at *IPI, LUH (*Germany) 2016
- Scholarship from *CNPq* for the Ph.D. program at *PUC-Rio* 2014-2018
- Scholarship *Bolsa Nota 10* from *FAPERJ* 2013
- Scholarship from *CAPES* for the M.Sc. program at *PUC-Rio* 2012

Invited Talks

- *Interpretando el mundo a través de imágenes y deep learning*, XI Electronic Week International Conference, 2022.
- *Desafíos del aprendizaje profundo en la visión por computador: Introducción al aprendizaje profundo y aplicaciones en teledetección,* V International Conference on Systems Engineering, 2022.
- *Deep Learning*, Summer School – IBT TU Braunschweig, 2022.
- *Aplicaciones de Deep Learning en Procesamiento Digital de Imágenes: desde el fondo del océano hasta el espacio exterior*, Capitulo de Ingeniería Electrónica CIP Cusco, 2022.
- *Segmentación de tipos de cultivos agrícolas con herramientas de machine learning e imágenes de teledetección,* Pontificia Universidad Católica del Perú PUCP, 2021.
- *Reconocimiento de cultivos agrícolas en regiones tropicales usando secuencias de imágenes de teledetección de sensores activos y pasivos*, International Conference on Computer Systems and Sciences, 2020.

Key Skills and Experience

Programming:	Python, MATLAB, C++, C#, R, Bash Script
Frameworks:	TensorFlow, Keras, PyTorch
Version Control:	Git, GitLab, GitHub
Containerization Tools:	Docker, Singularity
Software:	QGIS, ESA SNAP, MS Office

Languages

Spanish (native), English (advanced), Portuguese (intermediate)

Students

Co-advisor, M.Sc., Debwashis Borman (August 2026)
Co-advisor, M.Sc., Aditya Murti (May 2025)
Co-advisor, M.Sc., Eslam Nagah Sayed Mohamed Sharaawy (March 2025)
Co-advisor, M.Sc., Reiko Lettmoden (December 2024)
Co-advisor, M.Sc., Jose Manuel Gutierrez Castellanos (December 2024)
Co-advisor, M.Sc., Friedrich Hellweg (September 2024)
Co-advisor, M.Sc., David Hunkemöller (March 2024)
Co-advisor, M.Sc., William Alberto Ramirez Ruiz (April 2021)

Publications

Achanccaray Diaz, P., Wesche, L., Gerke, M., Hoyer, S., and Thiele, K. (2026). *Synthetic data generation for architectural typology documentation using diffusion models*, Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci., XLIX-B2-2026, 1381–1388, DOI:10.5194/isprs-archives-XLIX-B2-2026-1381-2026.

Saba, F., **Achanccaray Diaz, P.**, and Gerke, M. (2026). *Assessment of Different Architectures based on 2D-UNet, 3D-UNet and UNet-ConvLSTM for Land Use Land Cover Classification Using Multi-modal and Multi-temporal Satellite Images*, Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci., XLIX-B3-2026, 651–659, DOI:10.5194/isprs-archives-XLIX-B3-2026-651-2026.

Mawas, K., Maboudi, M., **Achanccaray, P.**, & Gerke, M. (2025). *Deep Filament Extraction for 3D Concrete Printing*. arXiv preprint arXiv:2512.00091.

Wesche, L.; **Achanccaray, P.**; Hoyer, S.; Unglaub, J.; Knufinke, U.; Gerke, M.; Krafczyk, C.; Thiele, K. (2025) *Konstruktionstypenvielfalt von Systemhallen in Stahlbauweise zum Ende der Hochmoderne*. Bautechnik102, H. 11, S. 645–655. DOI:10.1002/bate.70026

Fekete, S.P., **Achanccaray, P.M.**, Diaz, A., Figge, M., Fohrer, N., Gerke, M., Giutronich, S., Keldenich, P., Lutz, S., Perk, M., Reinhardt, A., Richter, C., Rieck, C., Riedel, B., Riedemann, T., Saba, F., Schrader, K., Schröter, A., Szafranski, D., Taghavi, A., Wagner, P.D., Wallner, T. (2025): *Wasserwirtschaft im Zeichen von Extremwetter*. Hydrologie & Wasserbewirtschaftung, 69, (6), 321-336. DOI: 10.5675/HyWa_2025.6_3

Wagner, P. D., Saba, F., **Achanccaray, P.**, and Fohrer, N.: *Drought-induced land use and land cover change impacts on hydrology: Insights from the Harz mountains, Germany*, EGUsphere [preprint], DOI:10.5194/egusphere-2025-3091, 2025.

Nyandwi, E., Gerke, M., **Achanccaray, P.**, & Lessmann, C. (2025). *Integrating remote sensing and deep learning for mapping urban housing wealth patterns*. Geo-Spatial Information Science, 1–19. DOI:10.1080/10095020.2025.2543496

R. Lettmoden, **P. M. Achanccaray Diaz**, K. Bittner, M. Gerke. *MultiGFM: Multi-temporal Framework for Multi-modal Geospatial Foundation Models*, Dreiländertagung der DGPF, der OVG und der SGPF, 2025, 182-200.

Saba, F., **Achanccaray, P.**, and Gerke, M. *Land use/Land cover mapping to monitor dead tree areas using multi-modal, multi-temporal Remote Sensing imagery and a deep learning model*, EGU General Assembly 2025, Vienna, Austria, EGU25-6990. DOI:10.5194/egusphere-egu25-6990

Rohrer M, Backhaus J, Bestmann U, De Arriba López V, **Achanccaray P**, Gerke M, et al. *Experimental studies on multi-scale data-driven methods within the framework of structural health monitoring*. Civil Engineering Design. 2025. DOI:10.1002/cend.202400036

Harb, S., **Achanccaray Diaz, P.**, Maboudi, M., and Gerke, M.: *Multi-temporal crack segmentation in concrete structures using deep learning approaches*, ISPRS Ann. Photogramm. Remote Sens. Spatial Inf. Sci., X-G-2025, 341–348, DOI:10.5194/isprs-annals-X-G-2025-341-2025, 2025.

Nyandwi, E., Gerke, M. & **Achanccaray, P**. *Local Evaluation of Large-scale Remote Sensing Machine Learning-generated Building and Road Dataset: The Case of Rwanda*. Journal of Photogrammetry, Remote Sensing and Geoinformation science – PFG (2024), DOI: 10.1007/s41064-024-00297-9

Heinrich, A., Mende, V., Wesche, L., & **Achanccaray, P**. (2024). *Database of recorded serial manufactured MLK-buildings (GDR) (Release 2) [Data set],* DOI:10.24355/dbbs.084-202403130624-0

Mawas, K., Maboudi, M., **Achanccaray, P.**, Gerke, M.: *Filament Extraction for 3D Printing*. In: Digital Concrete 2024 - Supplementary Proceedings. Braunschweig (2024)

A. Alamouri, J. Backhaus, V. De Arriba López, **P. M. Achanccaray Diaz**, M. Gerke. *High-Resolution Data Capture and Interpretation in Support of Port Infrastructure Maintenance*, DGPF-Jahrestagung, 2024.

M. Gerke, **P. M. Achanccaray Diaz**, S. Fekete, M. Figge, N. Fohrer, S. Giutronich, P. Keldenich, S. Lutz, M. Perk, A. Reinhardt, C. Richter, C. Rieck, B. Riedel, T. Riedemann, F. Saba, K. Schrader, A. Schröter, D. Szafranski, A. Taghavi, P. Wagner. *Extremwettermanagement mit digitalen Multiskalen-Methoden: Das EXDIMUM-Projekt*, DGPF-Jahrestagung, 2024.

De Arriba López, V., Maboudi, M., **Achanccaray, P.**, Gerke, M. *Automatic non-destructive UAV-based structural health monitoring of steel container cranes*. Applied Geomatics (2023), DOI: 10.1007/s12518-023-00542-7

Achanccaray, P., Gerke, M., Wesche, L., Hoyer, S., Thiele, K., Knufinke, U., and Krafczyk, C.: *On the assessment of instance segmentation for the automatic detection of specific constructions from very high resolution airborne imagery*, Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci., XLVIII-1/W2-2023, 1303–1309, DOI: 10.5194/isprs-archives-XLVIII-1-W2-2023-1303-2023, 2023

Wesche, L., **Achanccaray Diaz, P. M.**, Hoyer, S., Knufinke, U., Gerke, M., Krafczyk, C., & Thiele, K. (2023). *Dataset of german steel system halls from the period of high modernism [Data set]*. DOI: 10.24355/dbbs.084-202305261242-0

Leonhard Wesche, Sebastian Hoyer, Ulrich Knufinke, **Pedro Achanccaray**, Christina Krafczyk, Markus Gerke and Klaus Thiele. *Technologien für die Baudenkmalpflege: Erfassung und Analyse von Systemhallen der Hochmoderne*, in: Berichte zur Denkmalpflege in Niedersachsen 43 (2023), 2, pp. 61-65

Wesche, L., **Achanccaray, P.,** Hoyer, S. (2023). *Serielle Gebäude und wie man sie findet - Eine Methodik der Künstlichen Intelligenz zur Gebäudeerfassung*. In Gisbertz, O., Escherich, M., Hoyer, S., Putz, A., Weber, C. & DFG-Netzwerk Bauforschung Jüngere Baubestände 1945+ (Ed.). Reallabor Nachkriegsmoderne: Zum Umgang mit jüngeren Denkmalen. JOVIS Verlag GmbH

Achanccaray, P., Gerke, M., Wesche, L., Hoyer, S., Thiele, K., Knufinke, U., Krafczyk, C. *Automatic Detection of Specific Constructions on a Large Scale Using Deep Learning in Very High Resolution Airborne Imagery*.

Journal of Photogrammetry, Remote Sensing and Geoinformation science – PFG 91, pp. 189-209 (2023), DOI:10.1007/s41064-023-00237-z

Achanccaray, Pedro, Gerke, Markus, Hoyer, Sebastian, Knufinke, Ulrich, Krafczyk, Christina, Thiele, Klaus and Wesche, Leonhard. "*Deep Learning in der Denkmal-Inventarisation: Zur automatisierten luftbildbasierten Erfassung von Systembauwerken*" Die Denkmalpflege, 80, no. 2, 2022, pp. 162-16. DOI:10.1515/DKP-2022-2013

Heinrich, A., Mende, V., Wesche, L., & **Achanccaray, P.** (2022). *Database of recorded serial manufactured MLK-buildings (GDR) (Release 1) [Data set]*, DOI:10.24355/dbbs.084-202206080745-0

Ramirez, W., **Achanccaray, P.** & Pacheco, M.A. *A comparative study of Deep Learning architectures for Classification of Natural and Human-made Sea Events in SAR images*. Discov Artif Intell 2, 1 (2022). DOI:10.1007/s44163-022-00017-5

Bento, V., Kohler, M., **Diaz, P**. et al. *Improving deep learning performance by using Explainable Artificial Intelligence (XAI) approaches*. Discov Artif Intell 1, 9 (2021). DOI:10.1007/s44163-021-00008-y

Sanches, I. D., Feitosa, R. Q., Montibeller, B., **Achanccaray Diaz, P. M.**, Luiz, A. J. B., Soares, M. D., Prudente, V. H. R., Vieira, D. C., Maurano, L. E. P., Happ, P. N., Chamorro, J., and Oldoni, L. V.: *First results of the LEM benchmark database for agricultural applications*, Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci., XLIII-B5-2020, 251–256, DOI:10.5194/isprs-archives-XLIII-B5-2020-251-2020, 2020

Ramirez, W., **Achanccaray, P.**, Mendoza, L. F., and Pacheco, M. A. C.: *Deep convolutional neural networks for weed detection in agricultural crops using optical aerial images*, Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci., XLII-3/W12-2020, 551–555, DOI:10.5194/isprs-archives-XLII-3-W12-2020-551-2020, 2020

Soares, Marinalva Dias, Luciano Vieira Dutra, Gilson Alexandre Ostwald Pedro da Costa, Raul Queiroz Feitosa, Rogério Galante Negri, and **Pedro Diaz**. *A Meta-Methodology for Improving Land Cover and Land Use Classification with SAR Imagery*. Remote Sensing 12, no. 6 (2020): 961, DOI:10.3390/rs12060961

See the full list of my publications in my ResearchGate or Google Scholar profiles

Service

PFG Journal – *Editor Remote Sensing* (2026)
Mentor, *SISAY mentoring program* (2017, 2022-2024).
General Secretary, *IEEE Geoscience and Remote Sensing Society Brazil's Chapter* (2015-2016).
Co-organizer, *IEEE GRSS Young Professionals and ISPRS Summer School 2015*.
Reviewer of Journals: *IJRS, PFG, TGRS, J-STARS, GRSL.*